

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		16513	
Total Threats Mitigated	45	Total Packets Analyzed		4889211578	
High Priority	14	Malicious Packets		4871905914	
Low Priority	31	True Negatives Passed		12166	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		10	
ML Metrics	Value	SYN Flood		19	
False Positives	40	UDP Flood		6	
FP Rate	0.33%	Uncertain		10	
IF Precision	98.95%				
IF Recall	97.49%	Action		Count	
IF F1-Score	98.22%	Quarantined		10	
IF Accuracy	99.14%	Time Ban		10	
		Blackhole		0	
		Released		4	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	98.95%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.49%	Share of actual attacks successfully flagged
F1-Score	98.22%	Balanced measure combining Precision and Recall
Accuracy	99.14%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.33%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.51%	Actual attacks that went undetected
True Positive Rate (TPR)	97.49%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.67%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 3773	FN 97
Actual: Normal	FP 40	TN 12069

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	1239	0	0
Act: ICMP	0	406	0
Act: UDP	0	0	2128

2c. Hybrid Cascaded

Metric	Value	Description
Precision	98.95%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	97.49%	Share of actual attacks successfully flagged
F1-Score	98.22%	Balanced measure combining Precision and Recall
Accuracy	99.14%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.33%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	2.51%	Actual attacks that went undetected
True Positive Rate (TPR)	97.49%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.67%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 3773	FN 97
Actual: Normal	FP 40	TN 12069

2d. Response Latency

Metric	Value	Description
Detection Time	492.8 ms	Average time taken to detect an attack
Mitigation Response Time	15.5 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	3.49%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	11.64%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	5.71%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.17	4	3	2026-09-25 13:10:19	2026-09-25 13:16:16
10.0.0.18	3	3	2026-09-25 13:10:18	2026-09-25 13:13:29
10.0.0.19	3	3	2026-09-25 13:10:18	2026-09-25 13:13:29
10.0.0.24	3	3	2026-09-25 13:10:18	2026-09-25 13:16:20
10.0.0.25	3	3	2026-09-25 13:10:18	2026-09-25 13:16:20
10.0.0.16	3	2	2026-09-25 13:13:42	2026-09-25 13:16:15
10.0.0.21	2	1	2026-09-25 13:14:06	2026-09-25 13:16:21
10.0.0.20	1	1	2026-09-25 13:16:05	2026-09-25 13:16:16

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.22	1	1	2026-09-25 13:14:07	2026-09-25 13:14:17
10.0.0.23	1	1	2026-09-25 13:16:06	2026-09-25 13:16:16
10.0.0.2	1	1	2026-09-25 13:14:21	2026-09-25 13:14:31

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 13:10:18	10.0.0.25	DDoS	ICMP Flood	88.5%	High	Time Ban (1m)
2026-09-25 13:10:18	10.0.0.25	DDoS	ICMP Flood	88.5%	High	Time Ban
2026-09-25 13:10:18	10.0.0.24	DDoS	ICMP Flood	90.4%	High	Time Ban (1m)
2026-09-25 13:10:18	10.0.0.24	DDoS	ICMP Flood	90.4%	High	Time Ban
2026-09-25 13:10:18	10.0.0.19	DDoS	SYN Flood	91.4%	Critical	Quarantined
2026-09-25 13:10:18	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Quarantined
2026-09-25 13:10:19	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Quarantined
2026-09-25 13:10:29	10.0.0.19	DDoS	SYN Flood	91.4%	Critical	Time Ban (1m)
2026-09-25 13:10:29	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban (1m)
2026-09-25 13:10:29	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban (1m)
2026-09-25 13:11:19	10.0.0.25	DDoS	ICMP Flood	88.5%	High	Time Ban (2m)
2026-09-25 13:11:19	10.0.0.24	DDoS	ICMP Flood	90.4%	High	Time Ban (2m)
2026-09-25 13:11:29	10.0.0.19	DDoS	SYN Flood	91.4%	Critical	Time Ban (2m)
2026-09-25 13:11:29	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban (2m)
2026-09-25 13:11:29	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban (2m)
2026-09-25 13:13:19	10.0.0.25	DDoS	ICMP Flood	88.5%	High	Time Ban (5m)
2026-09-25 13:13:19	10.0.0.24	DDoS	ICMP Flood	90.4%	High	Time Ban (5m)
2026-09-25 13:13:29	10.0.0.19	DDoS	SYN Flood	91.4%	Critical	Time Ban (5m)
2026-09-25 13:13:29	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban (5m)
2026-09-25 13:13:29	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban (5m)
2026-09-25 13:13:35	10.0.0.17	DDoS	SYN Flood	92.7%	Critical	Time Ban
2026-09-25 13:13:35	10.0.0.19	DDoS	SYN Flood	91.4%	Critical	Time Ban
2026-09-25 13:13:35	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban
2026-09-25 13:13:42	10.0.0.16	DDoS	SYN Flood	90.4%	Critical	Quarantined
2026-09-25 13:13:53	10.0.0.16	DDoS	SYN Flood	91.4%	Critical	Time Ban (1m)
2026-09-25 13:13:53	10.0.0.16	DDoS	SYN Flood	91.4%	Critical	Time Ban
2026-09-25 13:14:06	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:14:07	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:14:14	10.0.0.20	DDoS	UDP Flood	77.5%	High	Time Ban (1m)
2026-09-25 13:14:14	10.0.0.20	DDoS	UDP Flood	77.5%	High	Time Ban
2026-09-25 13:14:16	10.0.0.21	DDoS	UDP Flood	86.5%	High	Time Ban (1m)
2026-09-25 13:14:16	10.0.0.21	DDoS	UDP Flood	86.5%	Critical	Time Ban
2026-09-25 13:14:17	10.0.0.22	DDoS	UDP Flood	86.2%	High	Time Ban (1m)
2026-09-25 13:14:18	10.0.0.22	DDoS	UDP Flood	86.2%	Critical	Time Ban
2026-09-25 13:14:31	10.0.0.2	DDoS	Uncertain	58.9%	Low	Released
2026-09-25 13:14:43	10.0.0.23	DDoS	ICMP Flood	96.6%	High	Time Ban (1m)
2026-09-25 13:14:43	10.0.0.23	DDoS	ICMP Flood	96.6%	High	Time Ban

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-25 13:14:53	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (2m)
2026-09-25 13:16:05	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:16:06	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:16:09	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:16:09	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-25 13:16:20	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-09-25 13:16:20	10.0.0.25	DDoS	Uncertain	-	Low	Released
2026-09-25 13:16:21	10.0.0.21	DDoS	Uncertain	-	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 13:16:21	Attack Stopped
10.0.0.25	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 13:16:20	Attack Stopped
10.0.0.24	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-25 13:16:20	Attack Stopped
10.0.0.23	ICMP Flood	0.6402	96.6%	High	Phase 2	1	10s	2026-09-25 13:16:16	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6359	90.1%	Critical	Phase 2	1	10s	2026-09-25 13:16:16	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6294	78.1%	High	Phase 2	1	11s	2026-09-25 13:16:16	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6471	93.4%	Critical	Phase 2	1	11s	2026-09-25 13:16:15	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6086	93.3%	Critical	Phase 2	2	1m 11s	2026-09-25 13:14:53	Attack persisted - escalating to time ban
10.0.0.2	Uncertain	0.6103	58.9%	Low	Phase 1	1	10s	2026-09-25 13:14:31	Attack Stopped
10.0.0.22	UDP Flood	0.6321	86.2%	High	Phase 2	1	10s	2026-09-25 13:14:17	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6472	86.5%	High	Phase 2	1	10s	2026-09-25 13:14:16	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6495	91.4%	Critical	Phase 2	1	11s	2026-09-25 13:13:53	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6395	92.7%	Critical	Phase 2	3	3m 10s	2026-09-25 13:13:29	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.18	SYN Flood	0.6250	91.4%	Critical	Phase 2	3	3m 11s	2026-09-25 13:13:29	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6236	91.4%	Critical	Phase 2	3	3m 11s	2026-09-25 13:13:29	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6459	90.4%	High	Phase 2	3	3m 1s	2026-09-25 13:13:19	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6354	88.5%	High	Phase 2	3	3m 1s	2026-09-25 13:13:19	Escalated to Time Ban
10.0.0.17	SYN Flood	0.6395	92.7%	Critical	Phase 2	2	1m 10s	2026-09-25 13:11:29	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6250	91.4%	Critical	Phase 2	2	1m 11s	2026-09-25 13:11:29	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6236	91.4%	Critical	Phase 2	2	1m 11s	2026-09-25 13:11:29	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6459	90.4%	High	Phase 2	2	1m 1s	2026-09-25 13:11:19	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6354	88.5%	High	Phase 2	2	1m 1s	2026-09-25 13:11:19	Escalated to Time Ban
10.0.0.17	SYN Flood	0.6395	92.7%	Critical	Phase 2	1	10s	2026-09-25 13:10:29	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6250	91.4%	Critical	Phase 2	1	11s	2026-09-25 13:10:29	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6236	91.4%	Critical	Phase 2	1	11s	2026-09-25 13:10:29	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE